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MIL-M-28787/525
15 SEPTEMBER 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY ADDRESS EXPANSION MODULE, KEY CODE DWF

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic memory address expansion module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.216 failures per million operating hours for class I or 8.127 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DWF.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 + 5°C 60 ∓ 5°C		Class II 4/ -55 + 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A5	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B7	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------- Module shall pass all remaining tests ----------						
D1	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D2,D3	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D4	Ibd	V	S	0.0	3.1	0.0	3.1	0.0	3.1	mA
D5-D7	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D8	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D9	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D10	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D11-D19	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D20	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D21-D32	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D33	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D34,D35	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D36-D43	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1,E2	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E3,E4	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E5	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E6	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E7-E39	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E40	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E41	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E42	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F2	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F3	Iih	VII	X	-0.3	0.04	-0.3	0.04	-0.3	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F4	Iih	VII	X	-1.5	0.14	-1.5	0.14	-1.5	0.14	mA
F5,F6	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F7	Iih	VII	X	-1.8	0.02	-1.8	0.02	-1.8	0.02	mA
F8	Iih	VII	X	-7.2	0.08	-7.2	0.08	-7.2	0.08	mA
F9	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F10	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F11-F17	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F18-F23	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F24,F25	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F26	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F27-F38	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F39	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F40,F41	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F42-F49	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G1	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G2	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G3	Iil	VIII	X	-0.3	0.04	-0.3	0.04	-0.3	0.04	mA
G4	Iil	VIII	X	-1.5	0.14	-1.5	0.14	-1.5	0.14	mA
G5,G6	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G7	Iil	VIII	X	-1.8	0.02	-1.8	0.02	-1.8	0.02	mA
G8	Iil	VIII	X	-7.2	0.08	-7.2	0.08	-7.2	0.08	mA
G9	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G10	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G11-G17	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G18-G23	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G24,G25	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G26	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G27-G38	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G39	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G40,G41	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G42-G49	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
H1-H42	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
J1-J42	Vol	X	X	0.0	0.3	0.0	0.5	0.0	0.5	V
K1-K30	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K31-K60	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L2	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L3	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L4-L8	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L9	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L10-L43	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1,M2	PSFC	XIII	S	6.3	11.7	6.3	11.7	6.3	11.7	uF
N	Icc	XIV	X	655	1572	655	1572	655	1572	mA
P1,P2	Functional	XV	X	See Table XV						
Q1,Q2	Dynamic	XVI	S	See Table XVI						
Q3,Q4	Dynamic	XVI	Q	See Table XVI						
R1	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R2,R3	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R4	Ci/o	XVIII	Q	4	150	4	150	4	150	pF
R5-R7	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R8	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10	Ci/o	XVIII	Q	4	100	4	100	4	100	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R11,R12	C I/O	XVIII	Q	2	75	2	75	2	75	pF
R13	C I/O	XVIII	Q	2	50	2	50	2	50	pF
R14-R19	C I/O	XVIII	Q	2	75	2	75	2	75	pF
R20	C I/O	XVIII	Q	6	75	6	75	6	75	pF
R21-R35	C I/O	XVIII	Q	2	75	2	75	2	75	pF
R36-R43	C I/O	XVIII	Q	1	75	1	75	1	75	pF
R44-RP1	C I/O	XVIII	Q	2	75	2	75	2	75	pF
S	Vic latchup	XIX	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTA	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/525-1 for class I and M28787/525-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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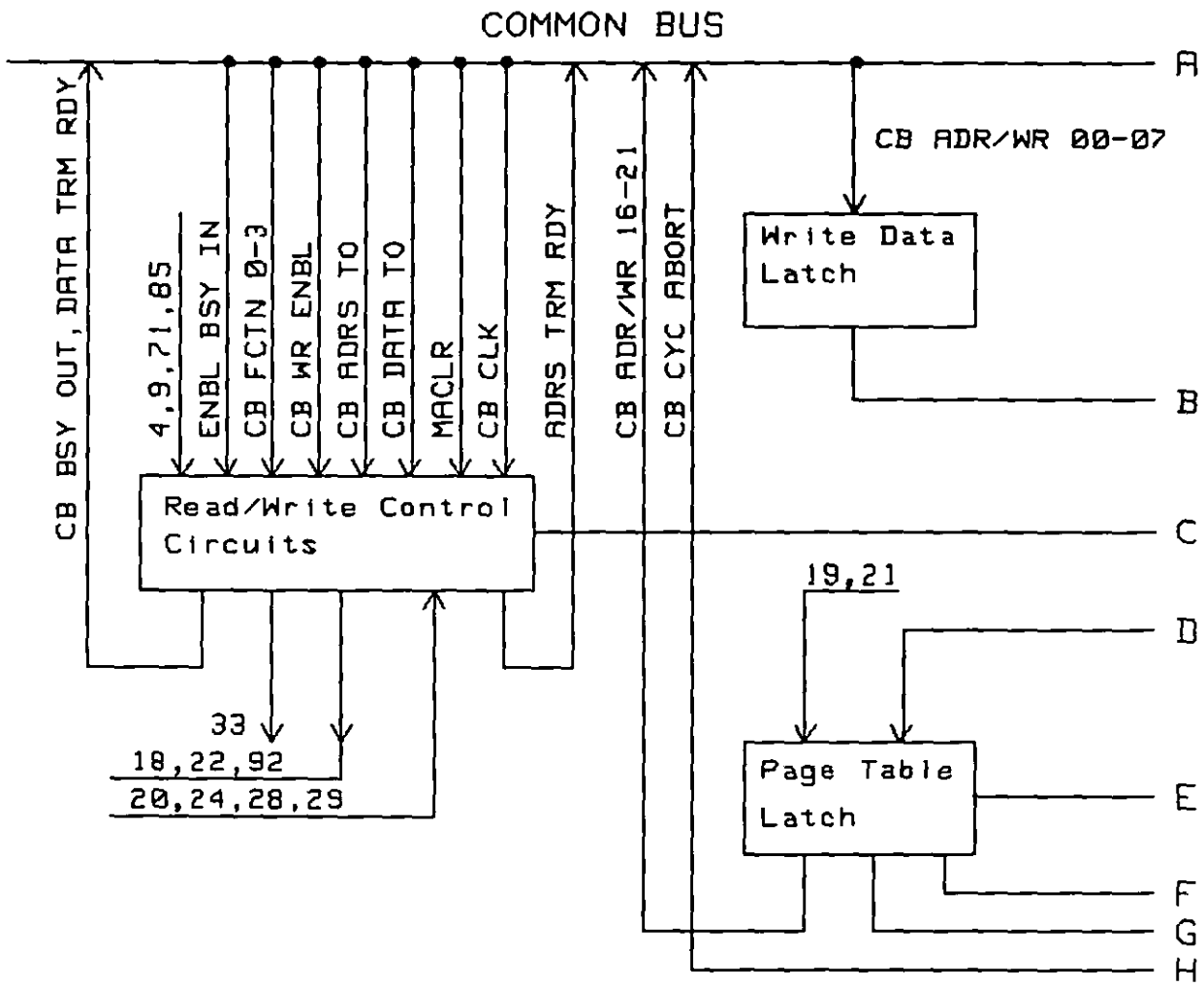


FIGURE 1. DWF functional block diagram.

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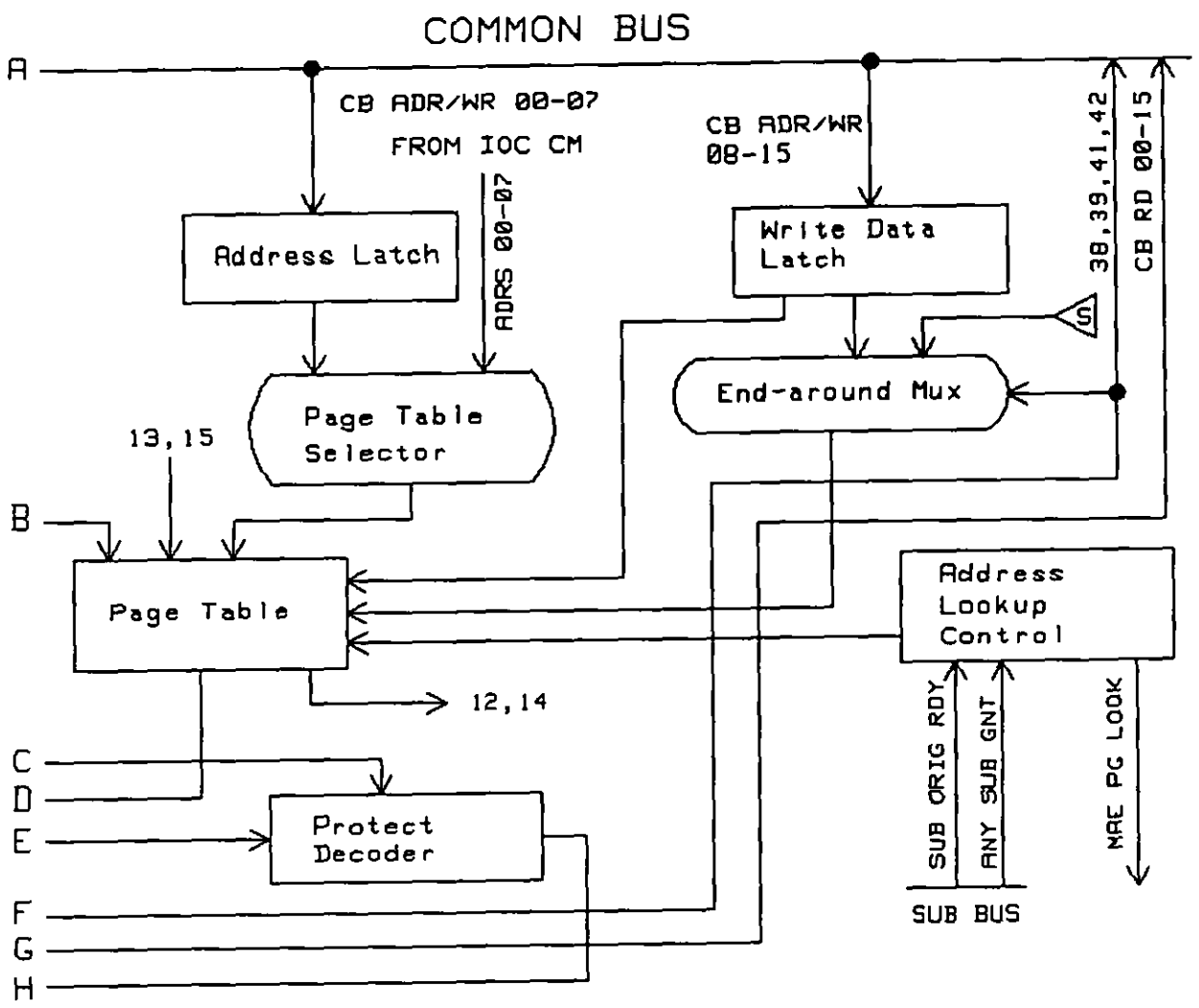


FIGURE 1. DWF functional block diagram - Continued.

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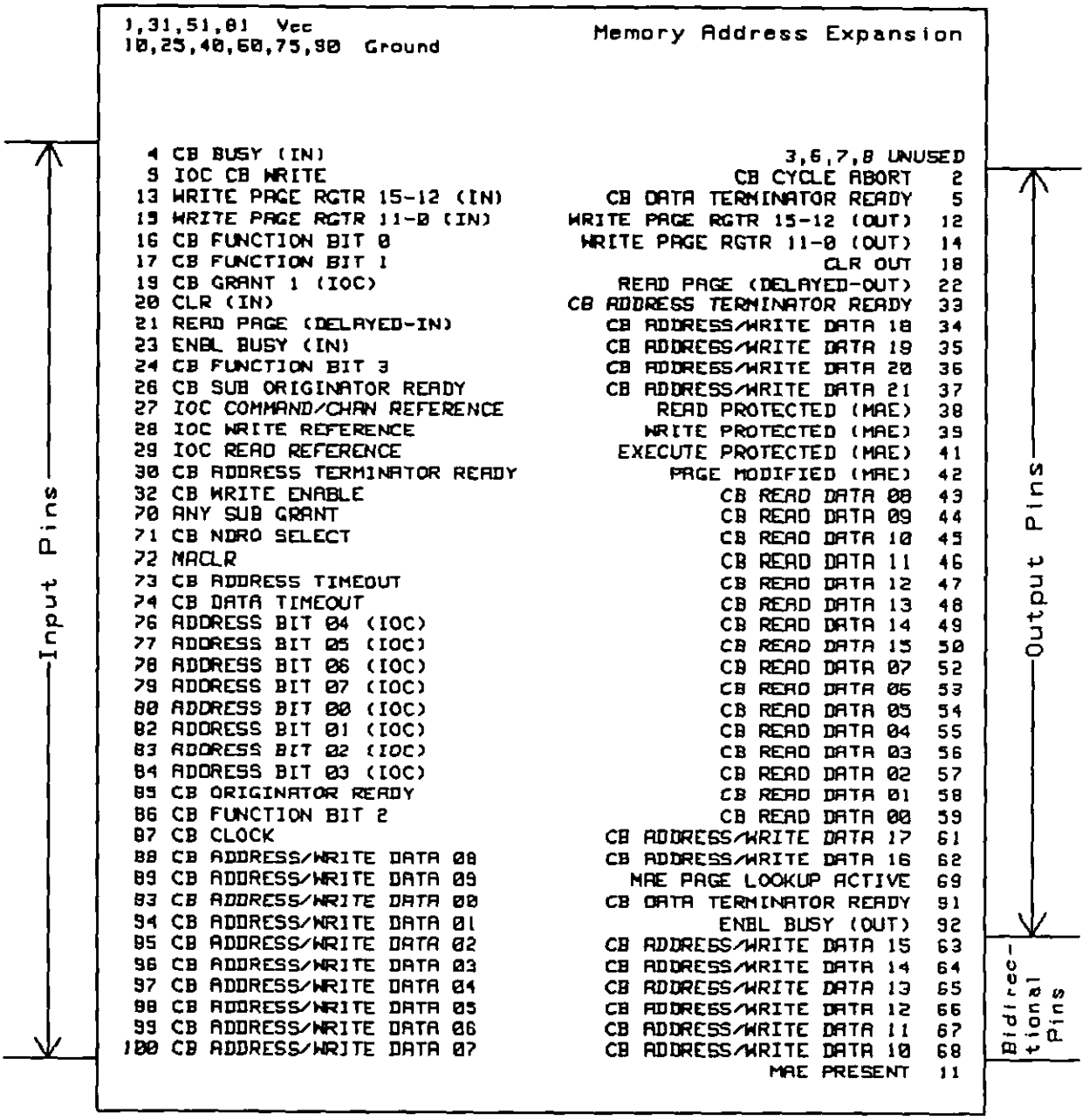


FIGURE 2. DWF pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L, and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L, and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

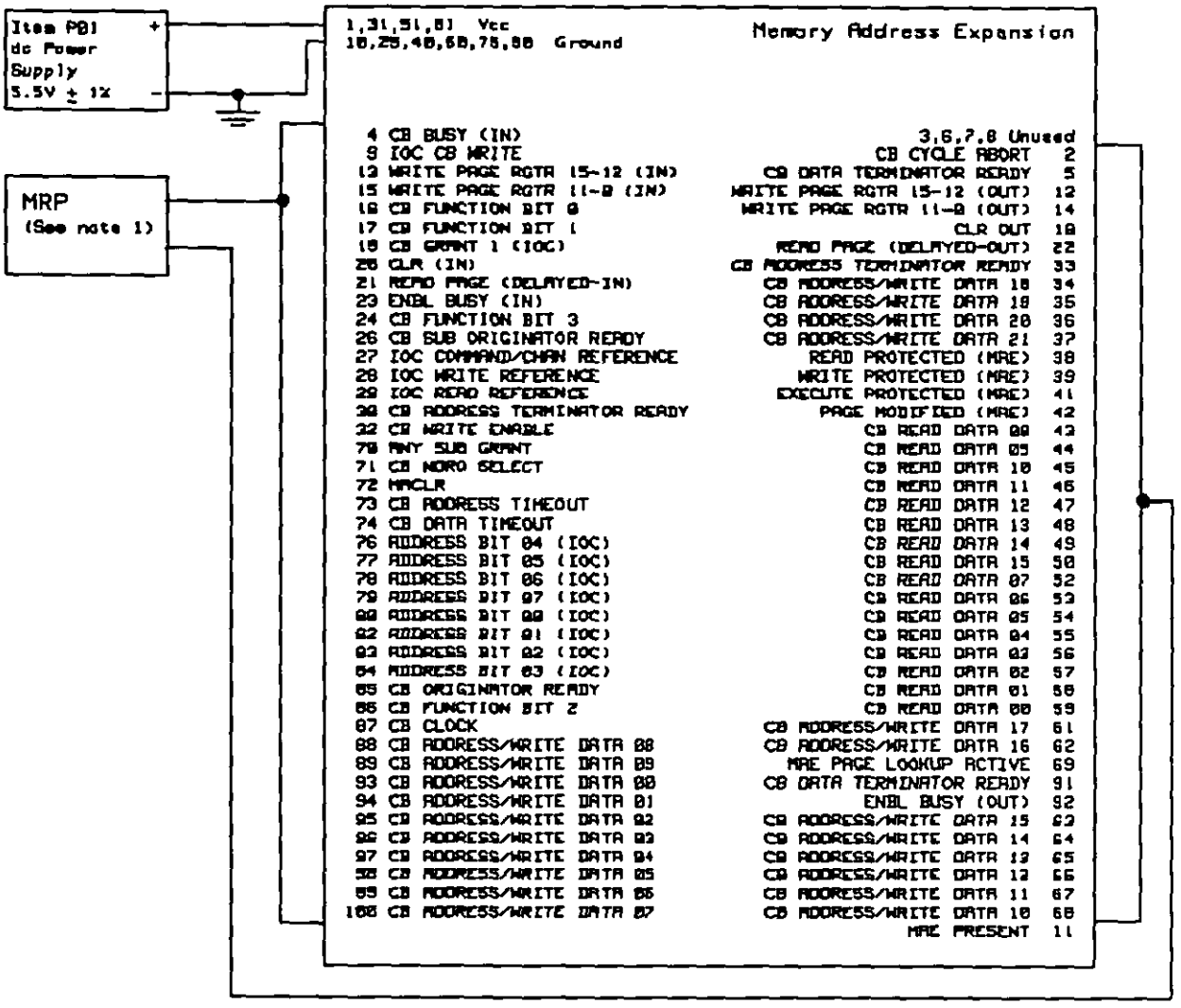
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. DWF operating temperature and component temperature test circuit.

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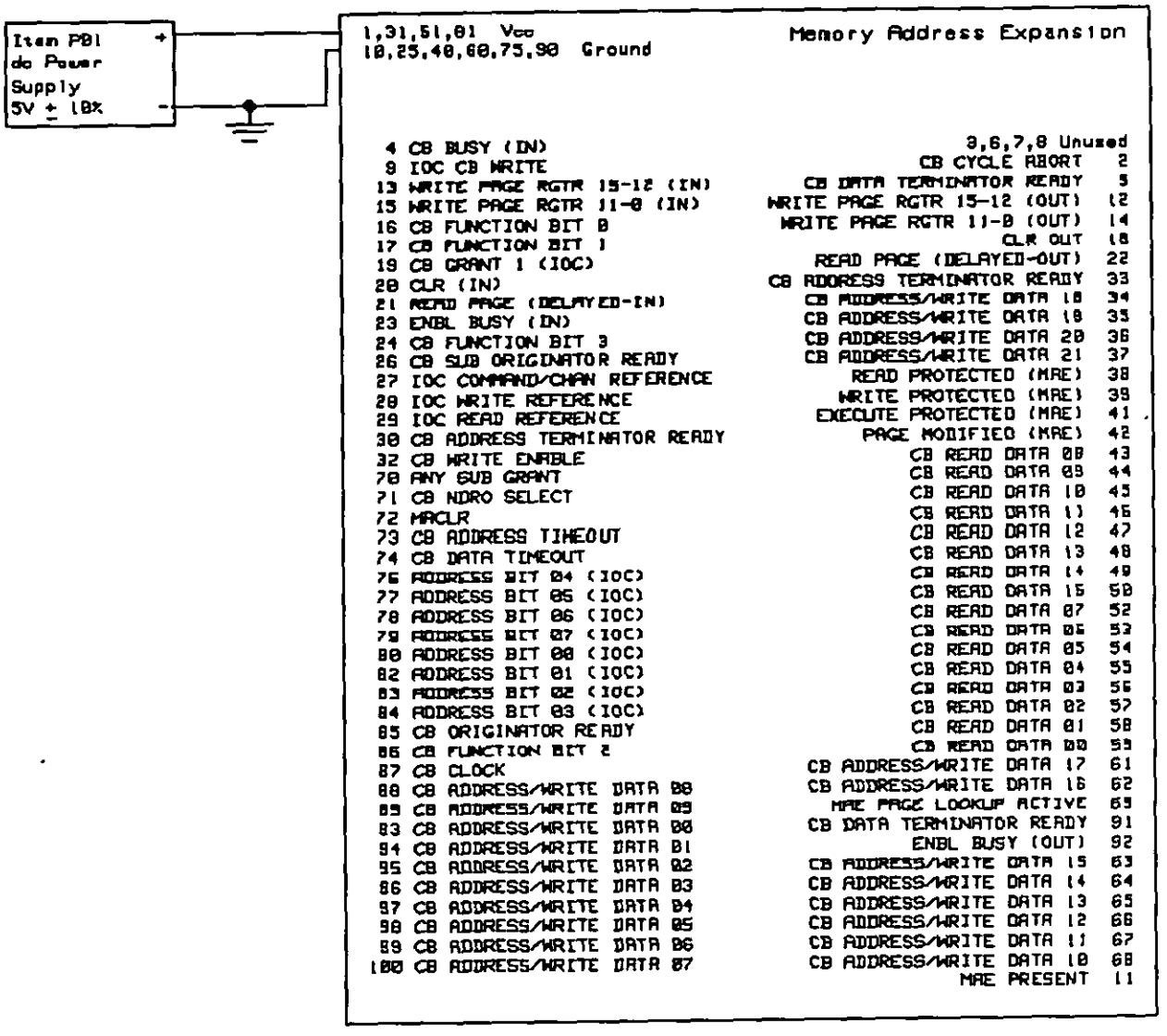


FIGURE 4. DWF vibration test circuit.

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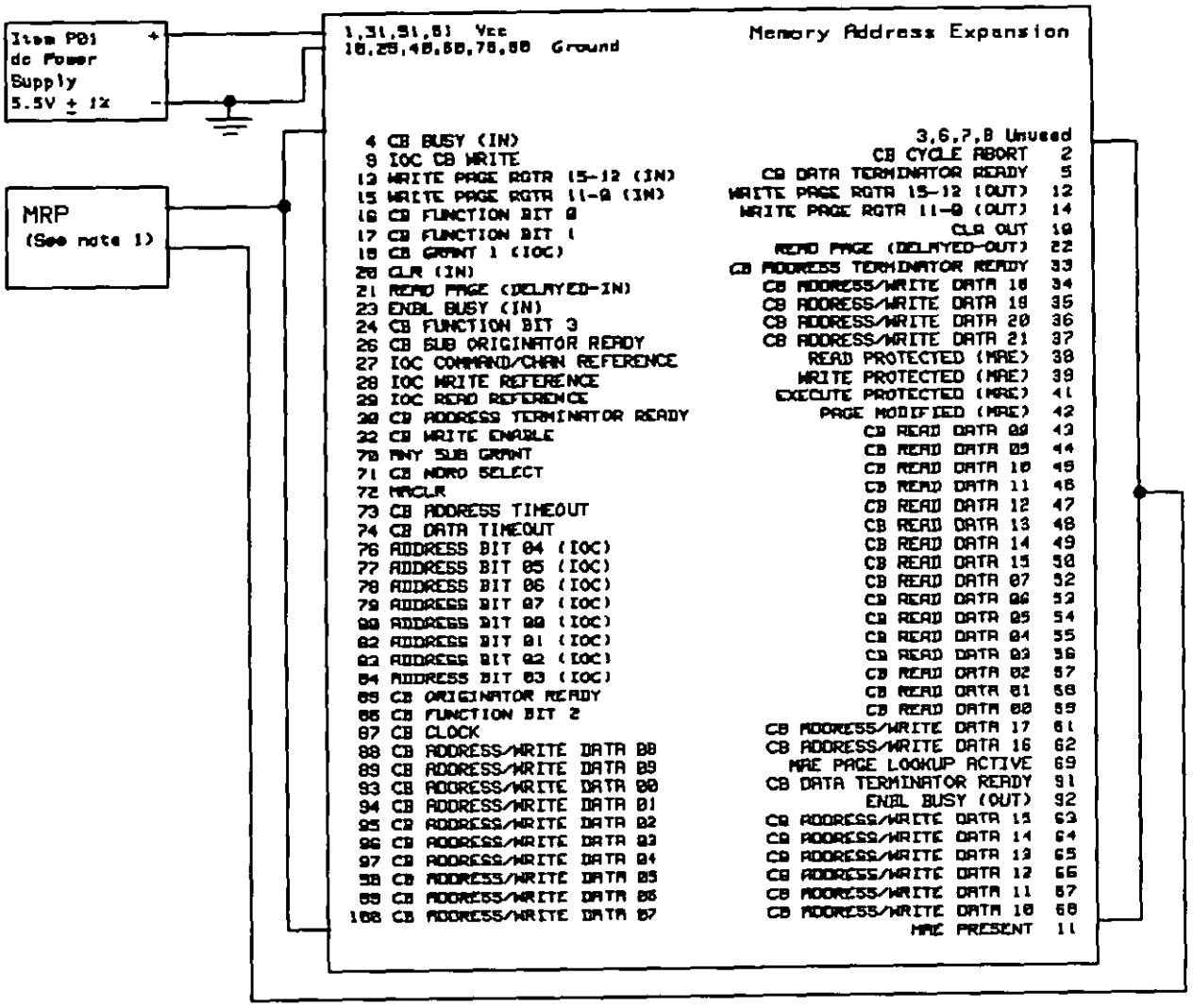
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. DWF life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 8.65 W (max); 5.80 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB BSY IN	-1.5	0.4	2.4	-0.2	0.02	0.1	100
IOC CB WR	-1.2	0.4	2.4	-2.0	0.05	1.0	75
WR 15-12 IN	-1.5	0.4	2.4	-0.3	0.04	1.0	75
WR 11-0 IN	-1.2	0.4	2.4	-1.5	0.14	3.1	150
CB FCTN 0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB FCTN 1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB GNT 1	-1.2	0.4	2.4	-1.8	0.02	0.1	75
CLR IN	-1.2	0.4	2.4	-7.2	0.08	0.4	125
RD PAGE IN	-1.5	0.4	2.4	-0.2	0.02	0.1	75
ENBL BSY IN	-1.2	0.4	2.4	-4.0	0.1	2.0	100
CB FCTN 3	-1.2	0.4	2.4	-0.6	0.02	0.1	75
SUB ORIG RDY	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOC COMM REF	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC WR REF	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOC RD REF	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS TRM RDY	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB WR ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ANY SUB GNT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB NDRO SEL	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MACLR	-1.2	0.4	2.4	-1.8	0.06	0.3	75
CB ADRS TO	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA TO	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS 04	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS 05	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS 06	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS 07	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ADRS 00	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS 01	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS 02	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADRS 03	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB ORIG RDY	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB FCTN 2	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	75
CB ADR/WR 08	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB ADR/WR 09	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB ADR/WR 00	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB ADR/WR 01	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB ADR/WR 02	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB ADR/WR 03	-1.2	0.4	2.4	-1.2	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB ADR/WR 04	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB ADR/WR 05	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB ADR/WR 06	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB ADR/WR 07	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB ADR/WR 15	NA	0.4	2.4	-0.65	0.07	NA	75
CB ADR/WR 14	NA	0.4	2.4	-0.65	0.07	NA	75
CB ADR/WR 13	NA	0.4	2.4	-0.65	0.07	NA	75
CB ADR/WR 12	NA	0.4	2.4	-0.65	0.07	NA	75
CB ADR/WR 11	NA	0.4	2.4	-0.65	0.07	NA	75
CB ADR/WR 10	NA	0.4	2.4	-0.65	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CB CYC ABORT	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
CB BSY OUT	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
WR 15-12 OUT	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
WR 11-0 OUT	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
CLR OUT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RD PAGE OUT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ADRS TRM RDY	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
CB ADR/WR 18	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB ADR/WR 19	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB ADR/WR 20	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB ADR/WR 21	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
RD PROTECTED	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
WR PROTECTED	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
EX PROTECTED	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
PAGE MOD	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
CB RD 08	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 09	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 10	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 11	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 12	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 13	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 14	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 15	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 07	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 06	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 05	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75

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6.3.2 Output parameters - Continued

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CB RD 04	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 03	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 02	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 01	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB RD 00	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB ADR/WR 17	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB ADR/WR 16	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
MAE PG LOOK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DATA TRM RDY	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ENBL BSY OUT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CB ADR/WR 15	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB ADR/WR 14	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB ADR/WR 13	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CB ADR/WR 12	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB ADR/WR 11	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB ADR/WR 10	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to ADRS TRM RDY	-	-	-	59	59	59	nS	12a
2	tpd	CB CLK to CB BSY OUT	-	-	-	104	104	104	nS	12a
3	tpd	CB CLK to DATA TRM RDY	-	-	-	104	104	104	nS	12a
4	tpd	CB GNT 1 to CB ADR/WR 18	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 19	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 20	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 21	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 17	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 16	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 15	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 14	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 13	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 12	-	-	-	49	49	49	nS	12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	CB GNT 1 to CB ADR/WR 11	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 10	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to RD PROTECTED	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to WR PROTECTED	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to EX PROTECTED	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to PAGE MOD	-	-	-	49	49	49	nS	12b
5	tsu	CB ORIG RDY to CB CLK	1	1	1	-	-	-	nS	12c
6	tsu	SUB ORIG RDY to CB CLK	18	18	18	-	-	-	nS	12c

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6.4 Module function. The Memory Address Expansion (MAE) module is employed by the Input/Output Controller (IOC) to expand its addressing range from 65,536 words to 4,194,304 words. Four sets of 64 page registers, each addressing 1024 words, are used to generate a 22-bit address that is sent to main memory. This process is identical to that which occurs in the MAE except that page set selection is accomplished by the page bits from the IOC control memory module. The MAE module contains the page table, a selector, address look-up control, and read/write control circuits. The page table is a 256 x 16 Random Access Memory (RAM) that contains 12 fixed base bits that form the upper 12 bits of the absolute address, a page modification bit that is set when the page has been written into and protect bits (read, write, and execute) which are applied to the protection check hardware and specify the protection checks to perform (a protect error results in a class II interrupt and an abort cycle). The page table selector selects either common bus address bits (in the case of a Central Processor (CP) write operation) or IOC MAE address bits for application to the address inputs of the page table. Since there are two copies of the page registers, one in the CP and one in the MAE module, the two sets must be kept identical. The CP performs read and write operations on the MAE page registers by non-memory data transfers on the commmon bus. When a load page instruction is executed, the data is loaded into the CP page register while the same data is transferred to the corresponding register in MAE. For store page instructions, the contents of the MAE register is read along with the CP page register. The contents of the two registers are read together and the result is stored. The read/write control circuits receive common bus control signals and generate the Address Terminator Ready (ATR) and Data Terminator Ready (DTR) signals to the master device. The address look-up control circuits receive sub-bus signals and generate the abort and let request signals in addition to supplying latch control signals for page table inputs.

6.4.1 Write data latches. Write data latches receive data information from the CP via the common bus when the load page macroinstruction is executed. The write clocking signal applied to the latches is provided by the read/write control circuits function decoder. Outputs from the latches are sent to the page table as data inputs.

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6.4.2 Address latch. The address latch receives address information from the CP (common bus bits 7-0) when the CP is performing a read or write of the page table. The MAE address clocking signal is a derivative of the function bits. Outputs of the address latch are supplied to a page table multiplexer which can select either a CP supplied address or address information from the IOC making a memory reference (read of a page table only). The output of the page table selector supplies page table address inputs.

6.4.3 Page table and latch. The page table contains four 256 x 4-bit RAMs. The page table is divided into four sets with each set containing 64 page registers which select pages with 1024 words per page. Page set selection is accomplished by the page bits from the IOC control memory module; page register selection is performed by an index which is the upper six bits of the relative address; a word within a page being selected by bits 9-0 of the relative address. Write enables are controlled by function bits, the address look-up control circuits, and the write MAE signal from the IOC common bus control module. Page table outputs are sent to the page table latches and drivers which can be either address bits 22-10 of the absolute address (in the case of a memory reference by an IOC) or data bits 15-0 (CP read of a page table). Also supplied with the absolute address bits are page register modified and execute, write, and read protection bits.

6.4.4 Protect decoder. The protect decoder circuit performs a memory protect check during address translation. Page table bits 14-12 are checked against protect conditions set by the IOC common bus control module. If a protect error occurs, a class II interrupt is generated and an abort cycle is started to inform the requestor and memory that a protect error has occurred.

6.4.5 End-around multiplexer. The end-around multiplexer selects either bits 15-12 of the common bus from the write data latch or protect bits read out of the page table or a memory reference. Also on a memory write reference the page modify bit (bit 15) is set.

6.4.6 Address look-up control circuits. The address look-up control circuits interface with the sub-bus and are employed during the sub-bus arbitration process. The IOC begins the request cycle by generating a sub request to the bus arbitrator. The bus arbitrator responds with a sub grant signal to the IOC and an any grant signal to the MAE. Address

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and page information is then placed on the address lines to MAE and the IOC generates the Sub Arbitrator Originator Ready (SA OR) signal to the MAE telling the MAE to begin address translation using the page table. The MAE also generates the let request signal to the bus arbitrator which is used to generate the main request to use the common bus.

6.4.7 Read/write control circuits. The read/write control circuits consist of a function decoder, ATR/DTR generator and clock and control logic. The read/write control circuits interface with the common bus and are employed in common bus transactions with the IOC and the CP. In the case of an IOC read or write of main memory, the main request for common bus usage will be generated by the bus arbitrator module which will send the IOC grant signal to the selected IOC and MAE. This signal generates an address enable to the common bus drivers to send the address and function to the slave device. Additionally, the IOC sends an OR signal to the MAE and the addressed device. Once the device has determined that it is being addressed, it generates a busy signal. The data will then be written into or read from main memory. In the case of a CP read or write into the page tables, the MAE module becomes a slave to the transaction. For a CP write, addresses are clocked in at the first OR and the function decoder generates the write signal which is used to clock in common bus data/address bits 15-0 into the page table. Additionally, the write signal is delayed to disable page table write enables after 100 nS. For a CP read, the write signal will be disabled and the data that is addressed will be read out of the page table and sent to the output latches and drivers. An ATR and DTR will be generated by MAE for the common bus transfers.

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6.4.8 Description of I/O pins.

Signal name	Pin number	I/O	Description
CB CYC ABORT	2	O	CB CYCLE ABORT
CB BSY IN	4	I	CB BUSY (IN)
CB BSY OUT	5	O	CB DATA TERMINATOR READY
IOC CB WR	9	I	IOC CB WRITE
WR 15-12 OUT	12	O	WRITE PAGE RGTR 15-12 (OUT)
WR 15-12 IN	13	I	WRITE PAGE RGTR 15-12 (IN)
WR 11-0 OUT	14	O	WRITE PAGE RGTR 11-0 (OUT)
WR 11-0 IN	15	I	WRITE PAGE RGTR 11-0 (IN)
CB FCTN 0	16	I	CB FUNCTION BIT 0
CB FCTN 1	17	I	CB FUNCTION BIT 1
CLR OUT	18	O	CLR OUT
CB GNT 1	19	I	CB GRANT 1 (IOC)
CLR IN	20	I	CLR (IN)
RD PAGE IN	21	I	READ PAGE (DELAYED-IN)
RD PAGE OUT	22	O	READ PAGE (DELAYED-OUT)
ENBL BSY IN	23	I	ENBL BUSY (IN)
CB FCTN 3	24	I	CB FUNCTION BIT 3
SUB ORIG RDY	26	I	CB SUB ORIGINATOR READY
IOC COMM REF	27	I	IOC COMMAND/CHAN REFERENCE
IOC WR REF	28	I	IOC WRITE REFERENCE

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC RD REF	29	I	IOC READ REFERENCE
ADRS TRM RDY	30	I	CB ADDRESS TERMINATOR READY
CB WR ENBL	32	I	CB WRITE ENABLE
ADRS TRM RDY	33	O	CB ADDRESS TERMINATOR READY
CB ADR/WR 18	34	O	CB ADDRESS/WRITE DATA 18
CB ADR/WR 19	35	O	CB ADDRESS/WRITE DATA 19
CB ADR/WR 20	36	O	CB ADDRESS/WRITE DATA 20
CB ADR/WR 21	37	O	CB ADDRESS/WRITE DATA 21
RD PROTECTED	38	O	READ PROTECTED (MAE)
WR PROTECTED	39	O	WRITE PROTECTED (MAE)
EX PROTECTED	41	O	EXECUTE PROTECTED (MAE)
PAGE MOD	42	O	PAGE MODIFIED (MAE)
CB RD 08	43	O	CB READ DATA 08
CB RD 09	44	O	CB READ DATA 09
CB RD 10	45	O	CB READ DATA 10
CB RD 11	46	O	CB READ DATA 11
CB RD 12	47	O	CB READ DATA 12
CB RD 13	48	O	CB READ DATA 13
CB RD 14	49	O	CB READ DATA 14
CB RD 15	50	O	CB READ DATA 15

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB RD 07	52	O	CB READ DATA 07
CB RD 06	53	O	CB READ DATA 06
CB RD 05	54	O	CB READ DATA 05
CB RD 04	55	O	CB READ DATA 04
CB RD 03	56	O	CB READ DATA 03
CB RD 02	57	O	CB READ DATA 02
CB RD 01	58	O	CB READ DATA 01
CB RD 00	59	O	CB READ DATA 00
CB ADR/WR 17	61	O	CB ADDRESS/WRITE DATA 17
CB ADR/WR 16	62	O	CB ADDRESS/WRITE DATA 16
CB ADR/WR 15	63	I/O	CB ADDRESS/WRITE DATA 15
CB ADR/WR 14	64	I/O	CB ADDRESS/WRITE DATA 14
CB ADR/WR 13	65	I/O	CB ADDRESS/WRITE DATA 13
CB ADR/WR 12	66	I/O	CB ADDRESS/WRITE DATA 12
CB ADR/WR 11	67	I/O	CB ADDRESS/WRITE DATA 11
CB ADR/WR 10	68	I/O	CB ADDRESS/WRITE DATA 10
MAE PG LOOK	69	O	MAE PAGE LOOKUP ACTIVE
ANY SUB GNT	70	I	ANY SUB GRANT
CB NDRO SEL	71	I	CB NDRO SELECT
MACLR	72	I	MACLR

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB ADRS TO	73	I	CB ADDRESS TIMEOUT
CB DATA TO	74	I	CB DATA TIMEOUT
ADRS 04	76	I	ADDRESS BIT 04 (IOC)
ADRS 05	77	I	ADDRESS BIT 05 (IOC)
ADRS 06	78	I	ADDRESS BIT 06 (IOC)
ADRS 07	79	I	ADDRESS BIT 07 (IOC)
ADRS 00	80	I	ADDRESS BIT 00 (IOC)
ADRS 01	82	I	ADDRESS BIT 01 (IOC)
ADRS 02	83	I	ADDRESS BIT 02 (IOC)
ADRS 03	84	I	ADDRESS BIT 03 (IOC)
CB ORIG RDY	85	I	CB ORIGINATOR READY
CB FCTN 2	86	I	CB FUNCTION BIT 2
CB CLK	87	I	CB CLOCK
CB ADR/WR 08	88	I	CB ADDRESS/WRITE DATA 08
CB ADR/WR 09	89	I	CB ADDRESS/WRITE DATA 09
DATA TRM RDY	91	O	CB DATA TERMINATOR READY
ENBL BSY OUT	92	O	ENBL BUSY (OUT)
CB ADR/WR 00	93	I	CB ADDRESS/WRITE DATA 00
CB ADR/WR 01	94	I	CB ADDRESS/WRITE DATA 01
CB ADR/WR 02	95	I	CB ADDRESS/WRITE DATA 02

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB ADR/WR 03	96	I	CB ADDRESS/WRITE DATA 03
CB ADR/WR 04	97	I	CB ADDRESS/WRITE DATA 04
CB ADR/WR 05	98	I	CB ADDRESS/WRITE DATA 05
CB ADR/WR 06	99	I	CB ADDRESS/WRITE DATA 06
CB ADR/WR 07	100	I	CB ADDRESS/WRITE DATA 07

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N740)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,3,6, 7,8	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	C	3,6,7,8				
A3	Isolation, unused to unused pins	3	6,7,8				
A4	Isolation, unused to unused pins	6	7,8				
A5	Isolation, unused to unused pins	7	8	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A5
Isolation measurement	NA	± 500 KOhm	A1-A5

2. The ohmmeter used shall be item R06.

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NOTES: - Continued.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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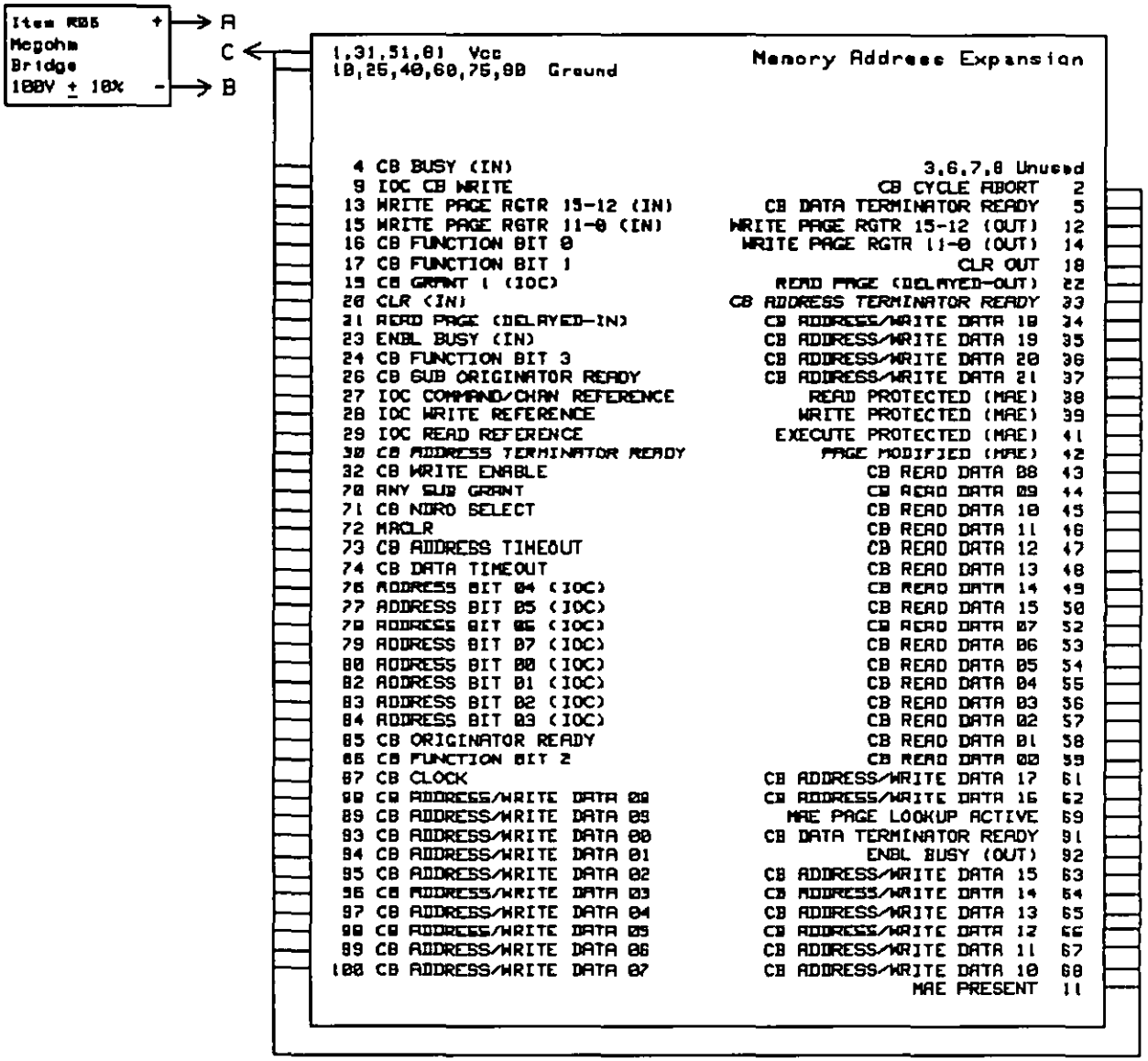


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81				
B7	Continuity, GND	10	11	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B7

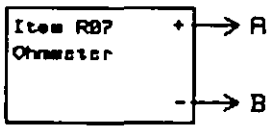
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 Vcc
10,25,40,60,75,90 Ground

Memory Address Expansion

4	CB BUSY (IN)	3,6,7,8	Unused
5	IOC CB WRITE	CB CYCLE ABORT	2
13	WRITE PAGE RGTR 15-12 (IN)	CB DATA TERMINATOR READY	5
15	WRITE PAGE RGTR 11-8 (IN)	WRITE PAGE RGTR 15-12 (OUT)	12
16	CB FUNCTION BIT 0	WRITE PAGE RGTR 11-8 (OUT)	14
17	CB FUNCTION BIT 1	CLR OUT	18
18	CB GRANT 1 (IOC)	READ PAGE (DELAYED-OUT)	22
20	CLR (IN)	CB ADDRESS TERMINATOR READY	33
21	READ PAGE (DELAYED-IN)	CB ADDRESS/WRITE DATA 18	34
23	ENBL BUSY (IN)	CB ADDRESS/WRITE DATA 19	35
24	CB FUNCTION BIT 3	CB ADDRESS/WRITE DATA 20	36
26	CB SUB ORIGINATOR READY	CB ADDRESS/WRITE DATA 21	37
27	IOC COMMAND/CHAN REFERENCE	READ PROTECTED (MAE)	38
28	IOC WRITE REFERENCE	WRITE PROTECTED (MAE)	39
29	IOC READ REFERENCE	EXECUTE PROTECTED (MAE)	41
30	CB ADDRESS TERMINATOR READY	PAGE MODIFIED (MAE)	42
32	CB WRITE ENABLE	CB READ DATA 08	43
70	ANY SUB GRANT	CB READ DATA 09	44
71	CB NDRO SELECT	CB READ DATA 10	45
72	MACLR	CB READ DATA 11	46
73	CB ADDRESS TIMEOUT	CB READ DATA 12	47
74	CB DATA TIMEOUT	CB READ DATA 13	48
76	ADDRESS BIT 04 (IOC)	CB READ DATA 14	49
77	ADDRESS BIT 05 (IOC)	CB READ DATA 15	50
78	ADDRESS BIT 06 (IOC)	CB READ DATA 07	52
79	ADDRESS BIT 07 (IOC)	CB READ DATA 06	53
80	ADDRESS BIT 00 (IOC)	CB READ DATA 05	54
82	ADDRESS BIT 01 (IOC)	CB READ DATA 04	55
83	ADDRESS BIT 02 (IOC)	CB READ DATA 03	56
84	ADDRESS BIT 03 (IOC)	CB READ DATA 02	57
85	CB ORIGINATOR READY	CB READ DATA 01	58
86	CB FUNCTION BIT 2	CB READ DATA 00	59
87	CB CLOCK	CB ADDRESS/WRITE DATA 17	61
88	CB ADDRESS/WRITE DATA 08	CB ADDRESS/WRITE DATA 16	62
89	CB ADDRESS/WRITE DATA 09	MAE PAGE LOOKUP ACTIVE	69
93	CB ADDRESS/WRITE DATA 00	CB DATA TERMINATOR READY	91
94	CB ADDRESS/WRITE DATA 01	ENBL BUSY (OUT)	92
95	CB ADDRESS/WRITE DATA 02	CB ADDRESS/WRITE DATA 15	63
96	CB ADDRESS/WRITE DATA 03	CB ADDRESS/WRITE DATA 14	64
97	CB ADDRESS/WRITE DATA 04	CB ADDRESS/WRITE DATA 13	65
98	CB ADDRESS/WRITE DATA 05	CB ADDRESS/WRITE DATA 12	66
99	CB ADDRESS/WRITE DATA 06	CB ADDRESS/WRITE DATA 11	67
100	CB ADDRESS/WRITE DATA 07	CB ADDRESS/WRITE DATA 10	68
		MAE PRESENT	11

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.83 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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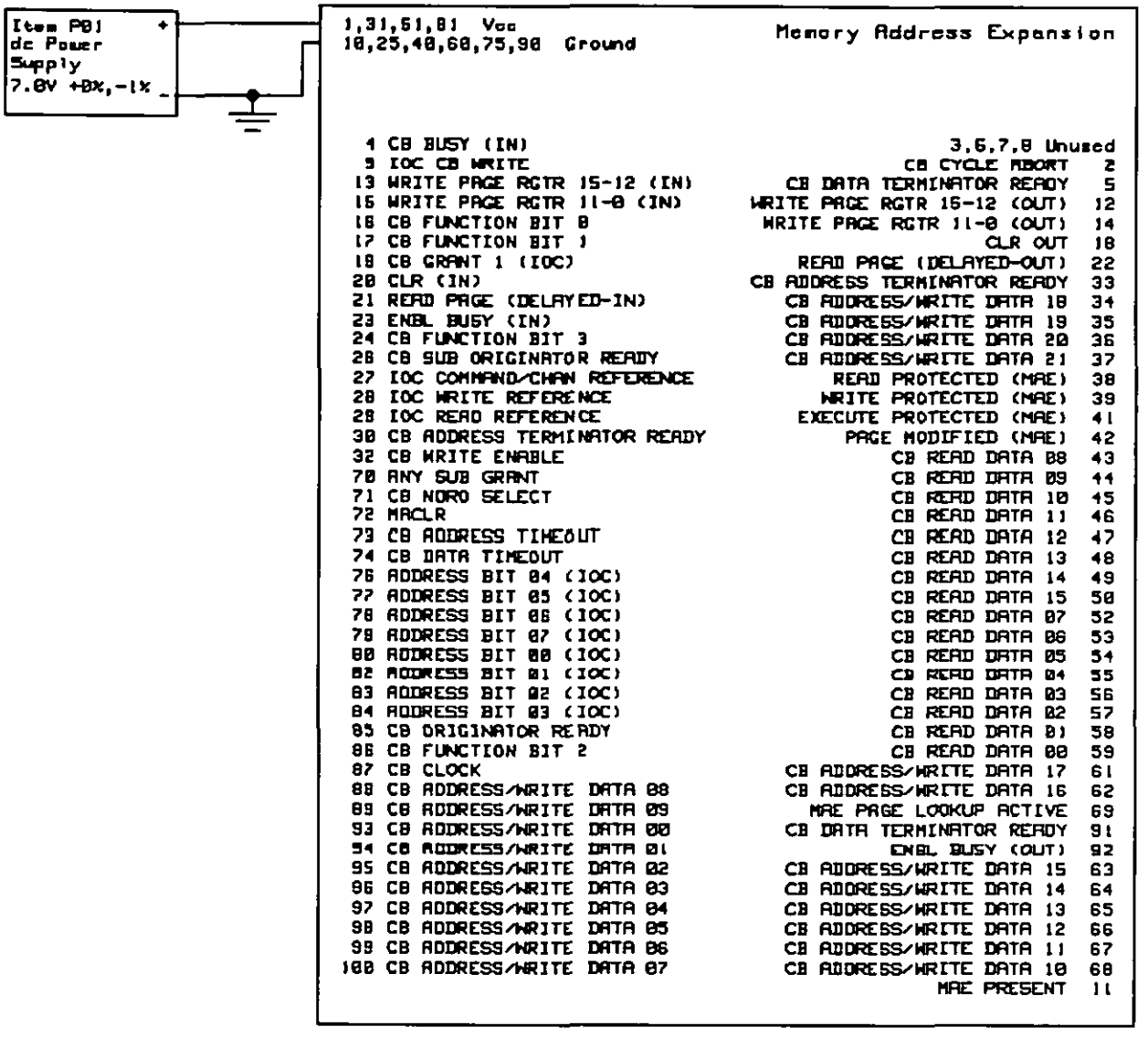


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	CB BSY IN	4	5.5	0.0	0.1	mA	1-3
D2	IOC CB WR	9	5.5	0.0	1.0	mA	1-3
D3	WR 15-12 IN	13	5.5	0.0	1.0	mA	1-3
D4	WR 11-0 IN	15	5.5	0.0	3.1	mA	1-3
D5	CB FCTN 0	16	5.5	0.0	0.1	mA	1-3
D6	CB FCTN 1	17					
D7	CB GNT 1	19	5.5	0.0	0.1	mA	1-3
D8	CLR IN	20	5.5	0.0	0.4	mA	1-3
D9	RD PAGE IN	21	5.5	0.0	0.1	mA	1-3
D10	ENBL BSY IN	23	5.5	0.0	2.0	mA	1-3
D11	CB FCTN 3	24	5.5	0.0	0.1	mA	1-3
D12	SUB ORIG RDY	26					
D13	IOC COMM REF	27					
D14	IOC WR REF	28					
D15	IOC RD REF	29					
D16	ADRS TRM RDY	30					
D17	CB WR ENBL	32					
D18	ANY SUB GNT	70					
D19	CB NDRO SEL	71	5.5	0.0	0.1	mA	1-3
D20	MACLR	72	5.5	0.0	0.3	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	CB ADRS TO	73	5.5	0.0	0.1	mA	1-3
D22	CB DATA TO	74					
D23	ADRS 04	76					
D24	ADRS 05	77					
D25	ADRS 06	78					
D26	ADRS 07	79					
D27	ADRS 00	80					
D28	ADRS 01	82					
D29	ADRS 02	83					
D30	ADRS 03	84					
D31	CB ORIG RDY	85					
D32	CB FCTN 2	86	5.5	0.0	0.1	mA	1-3
D33	CB CLK	87	5.5	0.0	1.0	mA	1-3
D34	CB ADR/WR 08	88	5.5	0.0	0.1	mA	1-3
D35	CB ADR/WR 09	89	5.5	0.0	0.1	mA	1-3
D36	CB ADR/WR 00	93	5.5	0.0	0.2	mA	1-3
D37	CB ADR/WR 01	94					
D38	CB ADR/WR 02	95					
D39	CB ADR/WR 03	96					
D40	CB ADR/WR 04	97	5.5	0.0	0.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	CB ADR/WR 05	98	5.5	0.0	0.2	mA	1-3
D42	CB ADR/WR 06	99					
D43	CB ADR/WR 07	100	5.5	0.0	0.2	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D43
Power supply current limit	2.36 A	NA	D1-D43
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D43
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D43 D1-D43
Input current measurement	NA NA NA NA NA NA NA NA	± 5 uA ± 50 uA ± 155 uA ± 5 uA ± 20 uA ± 5 uA ± 100 uA ± 5 uA	D1 D2,D3 D4 D5-D7 D8 D9 D10 D11-D19

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA NA NA NA NA	± 15 uA ± 5 uA ± 50 uA ± 5 uA ± 10 uA	D20 D21-D32 D33 D34,D35 D36-D43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	CB CYC ABORT	2	0.0	-325.0	-100.0	mA	1-4
E2	CB BSY OUT	5	0.0	-325.0	-100.0	mA	1-4
E3	WR 15-12 OUT	12	0.0	-100.0	-40.0	mA	1-4
E4	WR 11-0 OUT	14	0.0	-100.0	-40.0	mA	1-4
E5	CLR OUT	18	0.0	-150.0	-60.0	mA	1-4
E6	RD PAGE OUT	22	0.0	-260.0	-30.0	mA	1-4
E7	ADRS TRM RDY	33	0.0	-325.0	-100.0	mA	1-4
E8	CB ADR/WR 18	34					
E9	CB ADR/WR 19	35					
E10	CB ADR/WR 20	36					
E11	CB ADR/WR 21	37					
E12	RD PROTECTED	38					
E13	WR PROTECTED	39					
E14	EX PROTECTED	41					
E15	PAGE MOD	42					
E16	CB RD 08	43					
E17	CB RD 09	44					
E18	CB RD 10	45					
E19	CB RD 11	46					
E20	CB RD 12	47	0.0	-325.0	-100.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	CB RD 13	48	0.0	-325.0	-100.0	mA	1-4
E22	CB RD 14	49					
E23	CB RD 15	50					
E24	CB RD 07	52					
E25	CB RD 06	53					
E26	CB RD 05	54					
E27	CB RD 04	55					
E28	CB RD 03	56					
E29	CB RD 02	57					
E30	CB RD 01	58					
E31	CB RD 00	59					
E32	CB ADR/WR 17	61					
E33	CB ADR/WR 16	62					
E34	CB ADR/WR 15	63					
E35	CB ADR/WR 14	64					
E36	CB ADR/WR 13	65					
E37	CB ADR/WR 12	66					
E38	CB ADR/WR 11	67					
E39	CB ADR/WR 10	68	0.0	-325.0	-100.0	mA	1-4
E40	MAE PG LOOK	69	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	DATA TRM RDY	91	0.0	-325.0	-100.0	mA	1-4
E42	ENBL BSY OUT	92	0.0	-150.0	-60.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E42
Power supply current limit	2.36 A	NA	E1-E42
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E42
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E42 E1-E42
Output current measurement	NA NA NA NA NA NA NA NA	± 11.2 mA ± 3.0 mA ± 4.5 mA ± 11.5 mA ± 11.2 mA ± 4.5 mA ± 11.2 mA ± 4.5 mA	E1,E2 E3,E4 E5 E6 E7-E39 E40 E41 E42

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	CB BSY IN	4	2.4	-0.2	0.02	mA	1-3
F2	IOC CB WR	9	2.4	-2.0	0.05	mA	1-3
F3	WR 15-12 IN	13	2.4	-0.3	0.04	mA	1-3
F4	WR 11-0 IN	15	2.4	-1.5	0.14	mA	1-3
F5	CB FCTN 0	16	2.4	-0.6	0.02	mA	1-3
F6	CB FCTN 1	17	2.4	-0.6	0.02	mA	1-3
F7	CB GNT 1	19	2.4	-1.8	0.02	mA	1-3
F8	CLR IN	20	2.4	-7.2	0.08	mA	1-3
F9	RD PAGE IN	21	2.4	-0.2	0.02	mA	1-3
F10	ENBL BSY IN	23	2.4	-4.0	0.1	mA	1-3
F11	CB FCTN 3	24	2.4	-0.6	0.02	mA	1-3
F12	SUB ORIG RDY	26					
F13	IOC COMM REF	27					
F14	IOC WR REF	28					
F15	IOC RD REF	29					
F16	ADRS TRM RDY	30					
F17	CB WR ENBL	32	2.4	-0.6	0.02	mA	1-3
F18	CB ADR/WR 15	63	2.4	-0.65	0.07	mA	1-3
F19	CB ADR/WR 14	64					
F20	CB ADR/WR 13	65	2.4	-0.65	0.07	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	CB ADR/WR 12	66	2.4	-0.65	0.07	mA	1-3
F22	CB ADR/WR 11	67					
F23	CB ADR/WR 10	68	2.4	-0.65	0.07	mA	1-3
F24	ANY SUB GNT	70	2.4	-0.6	0.02	mA	1-3
F25	CB NDRO SEL	71	2.4	-0.6	0.02	mA	1-3
F26	MACLR	72	2.4	-1.8	0.06	mA	1-3
F27	CB ADRS TO	73	2.4	-0.6	0.02	mA	1-3
F28	CB DATA TO	74					
F29	ADRS 04	76					
F30	ADRS 05	77					
F31	ADRS 06	78					
F32	ADRS 07	79					
F33	ADRS 00	80					
F34	ADRS 01	82					
F35	ADRS 02	83					
F36	ADRS 03	84					
F37	CB ORIG RDY	85					
F38	CB FCTN 2	86	2.4	-0.6	0.02	mA	1-3
F39	CB CLK	87	2.4	-2.0	0.05	mA	1-3
F40	CB ADR/WR 08	88	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	CB ADR/WR 09	89	2.4	-0.6	0.02	mA	1-3
F42	CB ADR/WR 00	93	2.4	-1.2	0.04	mA	1-3
F43	CB ADR/WR 01	94					
F44	CB ADR/WR 02	95					
F45	CB ADR/WR 03	96					
F46	CB ADR/WR 04	97					
F47	CB ADR/WR 05	98					
F48	CB ADR/WR 06	99					
F49	CB ADR/WR 07	100	2.4	-1.2	0.04	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F49
Power supply current limit	2.36 A	NA	F1-F49
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F49

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F49
low, (VL)	0.0 V	± 100 mV	F1-F49
Input current measurement	NA	± 11 uA	F1
	NA	± 102 uA	F2
	NA	± 17 uA	F3
	NA	± 82 uA	F4
	NA	± 31 uA	F5,F6
	NA	± 91 uA	F7
	NA	± 360 uA	F8
	NA	± 11 uA	F9
	NA	± 205 uA	F10
	NA	± 31 uA	F11-F17
	NA	± 36 uA	F18-F23
	NA	± 31 uA	F24,F25
	NA	± 93 uA	F26
	NA	± 31 uA	F27-F38
	NA	± 102 uA	F39
	NA	± 31 uA	F40,F41
	NA	± 62 uA	F42-F49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CB BSY IN	4	0.4	-0.2	0.02	mA	1-3
G2	IOC CB WR	9	0.4	-2.0	0.05	mA	1-3
G3	WR 15-12 IN	13	0.4	-0.3	0.04	mA	1-3
G4	WR 11-0 IN	15	0.4	-1.5	0.14	mA	1-3
G5	CB FCTN 0	16	0.4	-0.6	0.02	mA	1-3
G6	CB FCTN 1	17	0.4	-0.6	0.02	mA	1-3
G7	CB GNT 1	19	0.4	-1.8	0.02	mA	1-3
G8	CLR IN	20	0.4	-7.2	0.08	mA	1-3
G9	RD PAGE IN	21	0.4	-0.2	0.02	mA	1-3
G10	ENBL BSY IN	23	0.4	-4.0	0.1	mA	1-3
G11	CB FCTN 3	24	0.4	-0.6	0.02	mA	1-3
G12	SUB ORIG RDY	26					
G13	IOC COMM REF	27					
G14	IOC WR REF	28					
G15	IOC RD REF	29					
G16	ADRS TRM RDY	30					
G17	CB WR ENBL	32	0.4	-0.6	0.02	mA	1-3
G18	CB ADR/WR 15	63	0.4	-0.65	0.07	mA	1-3
G19	CB ADR/WR 14	64					
G20	CB ADR/WR 13	65	0.4	-0.65	0.07	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	CB ADR/WR 12	66	0.4	-0.65	0.07	mA	1-3
G22	CB ADR/WR 11	67					
G23	CB ADR/WR 10	68	0.4	-0.65	0.07	mA	1-3
G24	ANY SUB GNT	70	0.4	-0.6	0.02	mA	1-3
G25	CB NDRO SEL	71	0.4	-0.6	0.02	mA	1-3
G26	MACLR	72	0.4	-1.8	0.06	mA	1-3
G27	CB ADRS TO	73	0.4	-0.6	0.02	mA	1-3
G28	CB DATA TO	74					
G29	ADRS 04	76					
G30	ADRS 05	77					
G31	ADRS 06	78					
G32	ADRS 07	79					
G33	ADRS 00	80					
G34	ADRS 01	82					
G35	ADRS 02	83					
G36	ADRS 03	84					
G37	CB ORIG RDY	85					
G38	CB FCTN 2	86	0.4	-0.6	0.02	mA	1-3
G39	CB CLK	87	0.4	-2.0	0.05	mA	1-3
G40	CB ADR/WR 08	88	0.4	-0.6	0.02	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:34Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	CB ADR/WR 09	89	0.4	-0.6	0.02	mA	1-3
G42	CB ADR/WR 00	93	0.4	-1.2	0.04	mA	1-3
G43	CB ADR/WR 01	94					
G44	CB ADR/WR 02	95					
G45	CB ADR/WR 03	96					
G46	CB ADR/WR 04	97					
G47	CB ADR/WR 05	98					
G48	CB ADR/WR 06	99					
G49	CB ADR/WR 07	100	0.4	-1.2	0.04	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G49
Power supply current limit	2.36 A	NA	G1-G49
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G49

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:34Z
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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G49
low, (VL)	0.0 V	± 100 mV	G1-G49
Input current measurement	NA	± 11 uA	G1
	NA	± 102 uA	G2
	NA	± 17 uA	G3
	NA	± 82 uA	G4
	NA	± 31 uA	G5,G6
	NA	± 91 uA	G7
	NA	± 360 uA	G8
	NA	± 11 uA	G9
	NA	± 205 uA	G10
	NA	± 31 uA	G11-G17
	NA	± 36 uA	G18-G23
	NA	± 31 uA	G24,G25
	NA	± 93 uA	G26
	NA	± 31 uA	G27-G38
	NA	± 102 uA	G39
	NA	± 31 uA	G40,G41
	NA	± 62 uA	G42-G49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	CB CYC ABORT	2	-0.4	2.4	4.5	V	1-4
H2	CB BSY OUT	5					
H3	WR 15-12 OUT	12					
H4	WR 11-0 OUT	14					
H5	CLR OUT	18	-0.4	2.4	4.5	V	1-4
H6	RD PAGE OUT	22	-0.16	2.4	4.5	V	1-4
H7	ADRS TRM RDY	33	-0.4	2.4	4.5	V	1-4
H8	CB ADR/WR 18	34					
H9	CB ADR/WR 19	35					
H10	CB ADR/WR 20	36					
H11	CB ADR/WR 21	37					
H12	RD PROTECTED	38					
H13	WR PROTECTED	39					
H14	EX PROTECTED	41					
H15	PAGE MOD	42					
H16	CB RD 08	43					
H17	CB RD 09	44					
H18	CB RD 10	45					
H19	CB RD 11	46					
H20	CB RD 12	47	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	CB RD 13	48	-0.4	2.4	4.5	V	1-4
H22	CB RD 14	49					
H23	CB RD 15	50					
H24	CB RD 07	52					
H25	CB RD 06	53					
H26	CB RD 05	54					
H27	CB RD 04	55					
H28	CB RD 03	56					
H29	CB RD 02	57					
H30	CB RD 01	58					
H31	CB RD 00	59					
H32	CB ADR/WR 17	61					
H33	CB ADR/WR 16	62					
H34	CB ADR/WR 15	63					
H35	CB ADR/WR 14	64					
H36	CB ADR/WR 13	65					
H37	CB ADR/WR 12	66					
H38	CB ADR/WR 11	67					
H39	CB ADR/WR 10	68					
H40	MAE PG LOOK	69	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	DATA TRM RDY	91	-0.4	2.4	4.5	V	1-4
H42	ENBL BSY OUT	92	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H42
Power supply current limit	2.36 A	NA	H1-H42
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA	H1-H5 H6 H7-H42
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H42 H1-H42
Output voltage measurement	NA	± 105 mV	H1-H42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

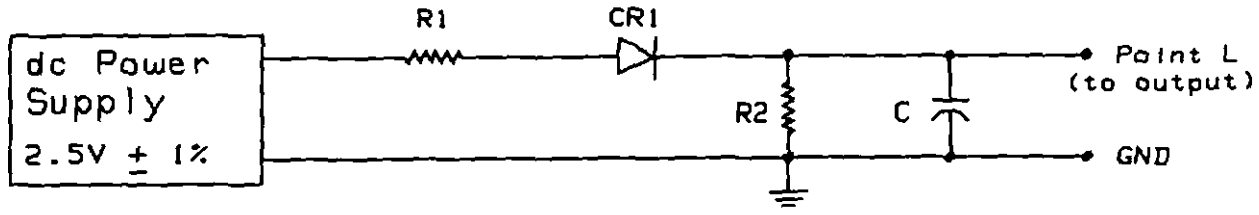
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,5,12,14,18,33, 34,35,36,37,38,39, 41,42,43,44,45,46, 47,48,49,50,52,53, 54,55,56,57,58,59, 61,62,63,64,65,66, 67,68,69,91,92	6200	6.0	0.5
22	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	CB CYC ABORT	2	6.0	0.0	0.5	V	1-4
J2	CB BSY OUT	5					
J3	WR 15-12 OUT	12					
J4	WR 11-0 OUT	14					
J5	CLR OUT	18	6.0	0.0	0.5	V	1-4
J6	RD PAGE OUT	22	1.6	0.0	0.5	V	1-4
J7	ADRS TRM RDY	33	6.0	0.0	0.5	V	1-4
J8	CB ADR/WR 18	34					
J9	CB ADR/WR 19	35					
J10	CB ADR/WR 20	36					
J11	CB ADR/WR 21	37					
J12	RD PROTECTED	38					
J13	WR PROTECTED	39					
J14	EX PROTECTED	41					
J15	PAGE MOD	42					
J16	CB RD 08	43					
J17	CB RD 09	44					
J18	CB RD 10	45					
J19	CB RD 11	46					
J20	CB RD 12	47	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	CB RD 13	48	6.0	0.0	0.5	V	1-4
J22	CB RD 14	49					
J23	CB RD 15	50					
J24	CB RD 07	52					
J25	CB RD 06	53					
J26	CB RD 05	54					
J27	CB RD 04	55					
J28	CB RD 03	56					
J29	CB RD 02	57					
J30	CB RD 01	58					
J31	CB RD 00	59					
J32	CB ADR/WR 17	61					
J33	CB ADR/WR 16	62					
J34	CB ADR/WR 15	63					
J35	CB ADR/WR 14	64					
J36	CB ADR/WR 13	65					
J37	CB ADR/WR 12	66					
J38	CB ADR/WR 11	67					
J39	CB ADR/WR 10	68					
J40	MAE PG LOOK	69	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	DATA TRM RDY	91	6.0	0.0	0.5	V	1-4
J42	ENBL BSY OUT	92	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J42
Power supply current limit	2.36 A	NA	J1-J42
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA	J1-J5 J6 J7-J42
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J42 J1-J42
Output voltage measurement	NA	± 25 mV	J1-J42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	CB CYC ABORT	2	2.4	-0.05	0.05	mA	1-3
K2	CB BSY OUT	5					
K3	ADRS TRM RDY	33					
K4	CB ADR/WR 18	34					
K5	CB ADR/WR 19	35					
K6	CB ADR/WR 20	36					
K7	CB ADR/WR 21	37					
K8	RD PROTECTED	38					
K9	WR PROTECTED	39					
K10	EX PROTECTED	41					
K11	PAGE MOD	42					
K12	CB RD 08	43					
K13	CB RD 09	44					
K14	CB RD 10	45					
K15	CB RD 11	46					
K16	CB RD 12	47					
K17	CB RD 13	48					
K18	CB RD 14	49					
K19	CB RD 15	50					
K20	CB RD 07	52	2.4	-0.05	0.05	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K21	CB RD 06	53	2.4	-0.05	0.05	mA	1-3
K22	CB RD 05	54					
K23	CB RD 04	55					
K24	CB RD 03	56					
K25	CB RD 02	57					
K26	CB RD 01	58					
K27	CB RD 00	59					
K28	CB ADR/WR 17	61					
K29	CB ADR/WR 16	62					
K30	DATA TRM RDY	91	2.4	-0.05	0.05	mA	1-3
Low level							
K31	CB CYC ABORT	2	0.4	-0.05	0.05	mA	1-3
K32	CB BSY OUT	5					
K33	ADRS TRM RDY	33					
K34	CB ADR/WR 18	34					
K35	CB ADR/WR 19	35					
K36	CB ADR/WR 20	36					
K37	CB ADR/WR 21	37					
K38	RD PROTECTED	38					
K39	WR PROTECTED	39	0.4	-0.05	0.05	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:34Z
Check the source to verify that this is the current version before use.

TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K40	EX PROTECTED	41	0.4	-0.05	0.05	mA	1-3
K41	PAGE MOD	42					
K42	CB RD 08	43					
K43	CB RD 09	44					
K44	CB RD 10	45					
K45	CB RD 11	46					
K46	CB RD 12	47					
K47	CB RD 13	48					
K48	CB RD 14	49					
K49	CB RD 15	50					
K50	CB RD 07	52					
K51	CB RD 06	53					
K52	CB RD 05	54					
K53	CB RD 04	55					
K54	CB RD 03	56					
K55	CB RD 02	57					
K56	CB RD 01	58					
K57	CB RD 00	59					
K58	CB ADR/WR 17	61					
K59	CB ADR/WR 16	62					
K60	DATA TRM RDY	91	0.4	-0.05	0.05	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:34Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K60
Power supply current limit	2.36 A	NA	K1-K60
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K30 K31-K60
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K60 K1-K60
Output current measurement	NA	± 5 uA	K1-K60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	CB BSY IN	4	-18.0	-1.5	0.0	V	1-3
L2	IOC CB WR	9	-18.0	-1.2	0.0	V	1-3
L3	WR 15-12 IN	13	-18.0	-1.5	0.0	V	1-3
L4	WR 11-0 IN	15	-18.0	-1.2	0.0	V	1-3
L5	CB FCTN 0	16					
L6	CB FCTN 1	17					
L7	CB GNT 1	19					
L8	CLR IN	20	-18.0	-1.2	0.0	V	1-3
L9	RD PAGE IN	21	-18.0	-1.5	0.0	V	1-3
L10	ENBL BSY IN	23	-18.0	-1.2	0.0	V	1-3
L11	CB FCTN 3	24					
L12	SUB ORIG RDY	26					
L13	IOC COMM REF	27					
L14	IOC WR REF	28					
L15	IOC RD REF	29					
L16	ADRS TRM RDY	30					
L17	CB WR ENBL	32					
L18	ANY SUB GNT	70					
L19	CB NDRO SEL	71					
L20	MACLR	72	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L21	CB ADRS TO	73	-18.0	-1.2	0.0	V	1-3
L22	CB DATA TO	74					
L23	ADRS 04	76					
L24	ADRS 05	77					
L25	ADRS 06	78					
L26	ADRS 07	79					
L27	ADRS 00	80					
L28	ADRS 01	82					
L29	ADRS 02	83					
L30	ADRS 03	84					
L31	CB ORIG RDY	85					
L32	CB FCTN 2	86					
L33	CB CLK	87					
L34	CB ADR/WR 08	88					
L35	CB ADR/WR 09	89					
L36	CB ADR/WR 00	93					
L37	CB ADR/WR 01	94					
L38	CB ADR/WR 02	95					
L39	CB ADR/WR 03	96					
L40	CB ADR/WR 04	97	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	CB ADR/WR 05	98	-18.0	-1.2	0.0	V	1-3
L42	CB ADR/WR 06	99	|	|	|	|	|
L43	CB ADR/WR 07	100	-18.0	-1.2	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L43
Power supply current limit	NA	NA	L1-L43
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L43
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L43 L1-L43
Input voltage measurement	 NA NA NA NA NA NA	 ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	 L1 L2 L3 L4-L8 L9 L10-L43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	6.3	11.7	uF	1-4
M2	PSFC	51	6.3	11.7	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA	± 270 nF	M1,M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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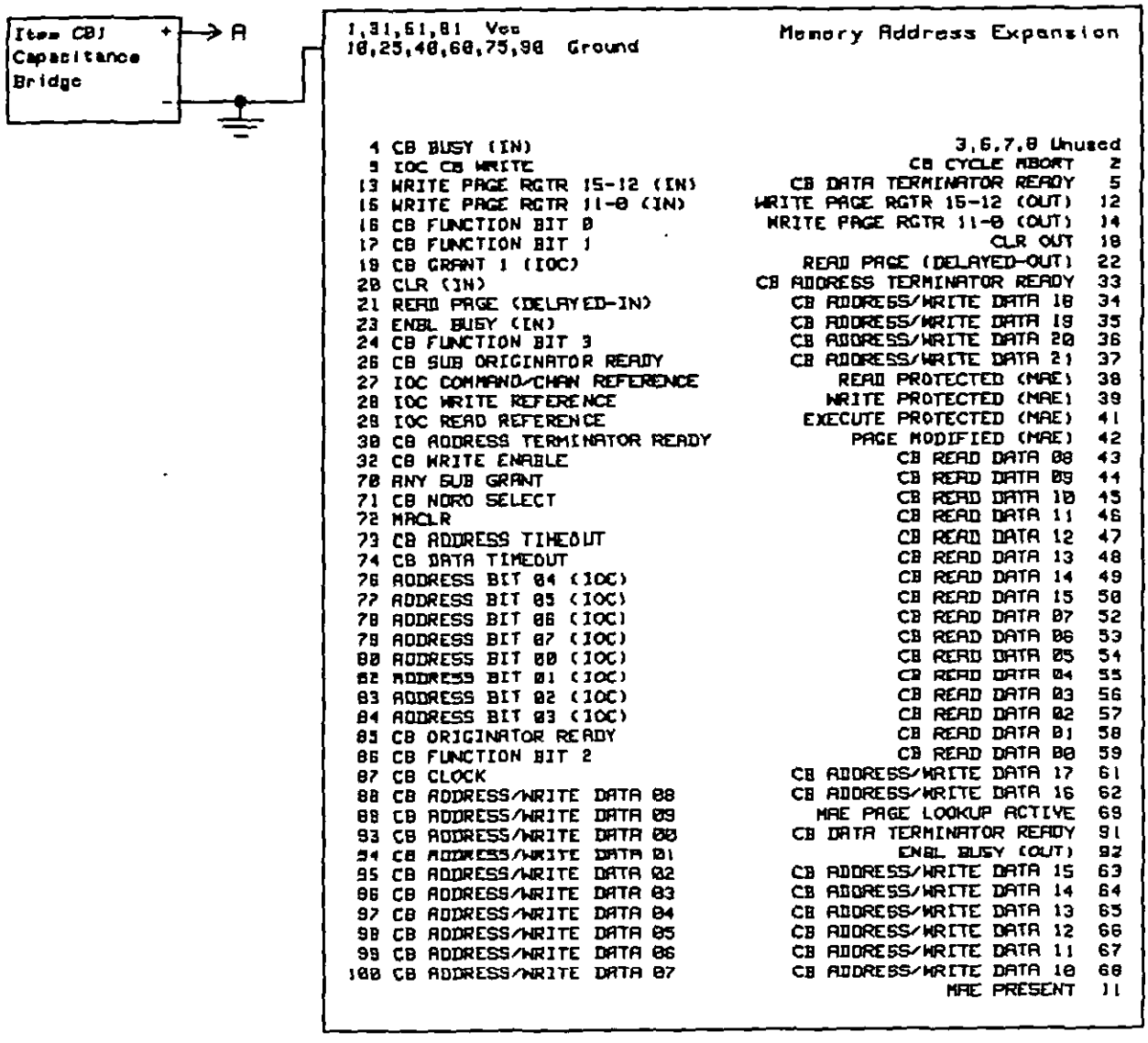


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	655	1572	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.36 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 45.8 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.36 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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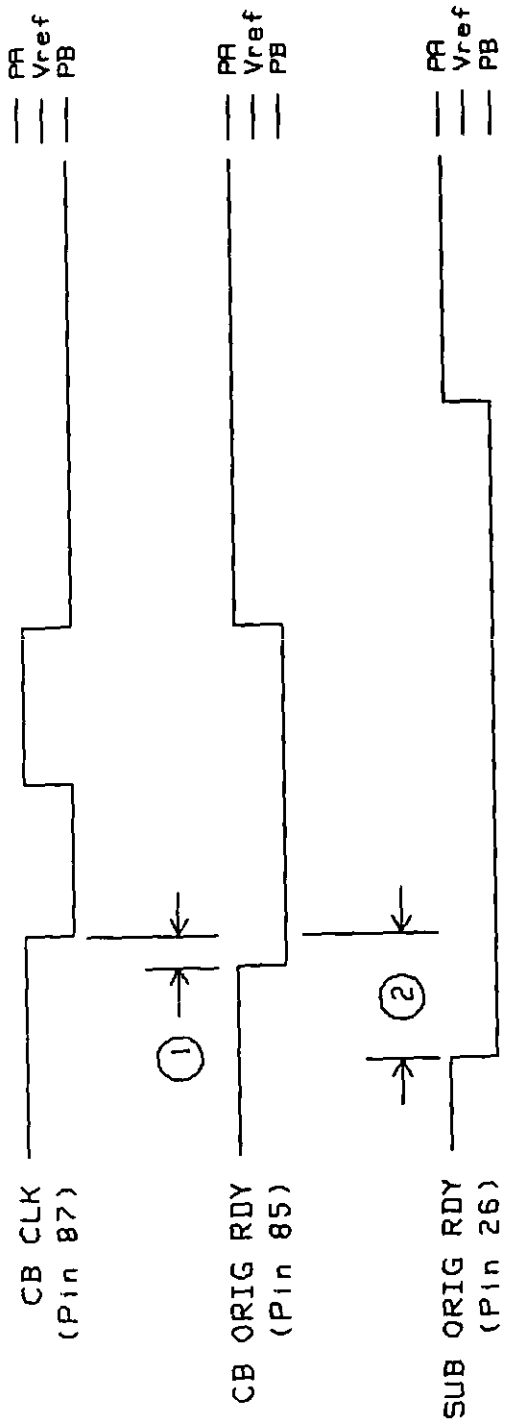


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) CB ORIG RDY to CB CLK. Setup time: 1 nS minimum.
2. (2) SUB ORIG RDY to CB CLK. Setup time: 18 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.36 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

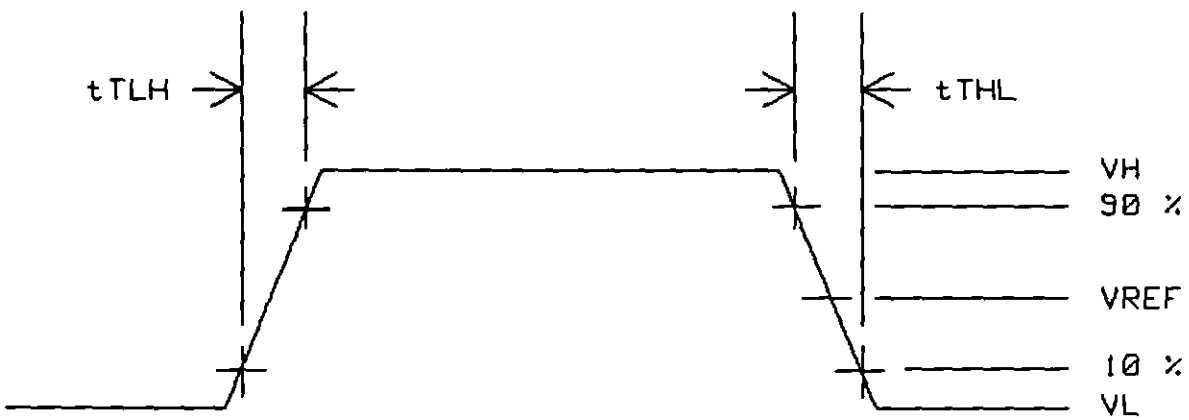
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to ADRS TRM RDY	-	-	-	59	59	59	nS	12a
2	tpd	CB CLK to CB BSY OUT	-	-	-	104	104	104	nS	12a
3	tpd	CB CLK to DATA TRM RDY	-	-	-	104	104	104	nS	12a
4	tpd	CB GNT 1 to CB ADR/WR 18	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 19	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 20	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 21	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 17	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 16	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 15	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 14	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 13	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 12	-	-	-	49	49	49	nS	12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	CB GNT 1 to CB ADR/WR 11	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to CB ADR/WR 10	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to RD PROTECTED	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to WR PROTECTED	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to EX PROTECTED	-	-	-	49	49	49	nS	12b
4	tpd	CB GNT 1 to PAGE MOD	-	-	-	49	49	49	nS	12b
5	tsu	CB ORIG RDY to CB CLK	1	1	1	-	-	-	nS	12c
6	tsu	SUB ORIG RDY to CB CLK	18	18	18	-	-	-	nS	12c

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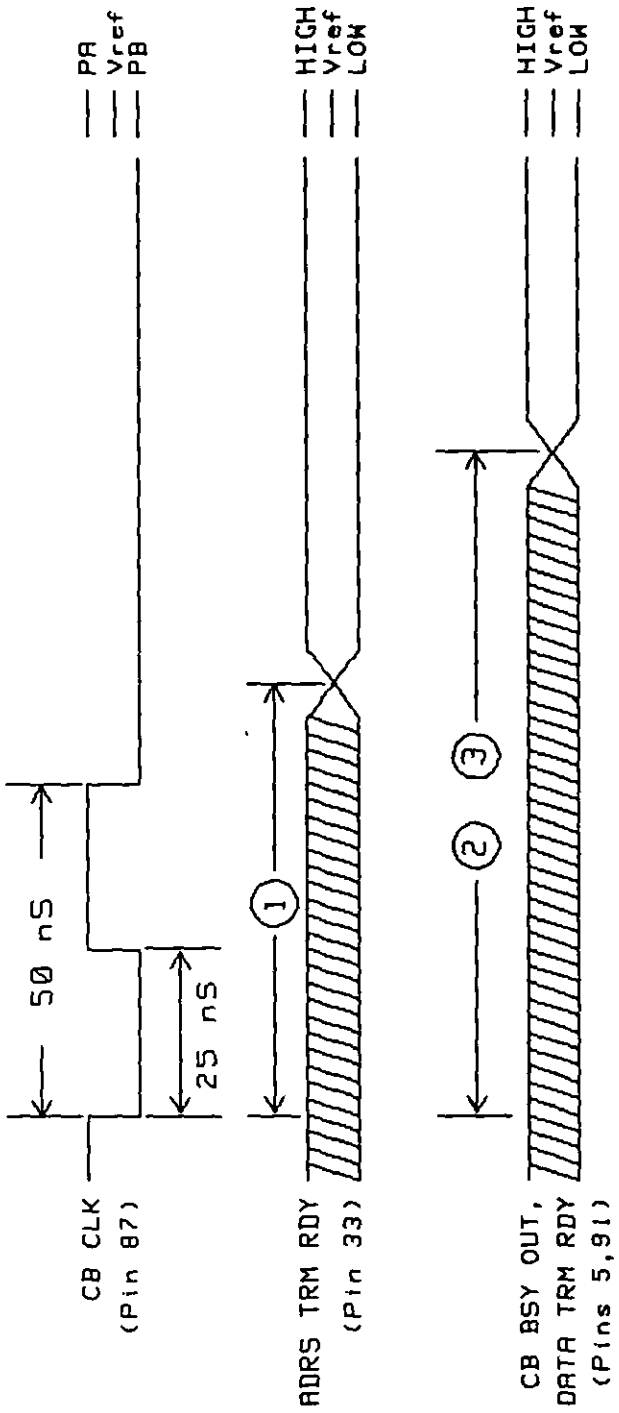


FIGURE 12a. Dynamic waveforms for table XVII.

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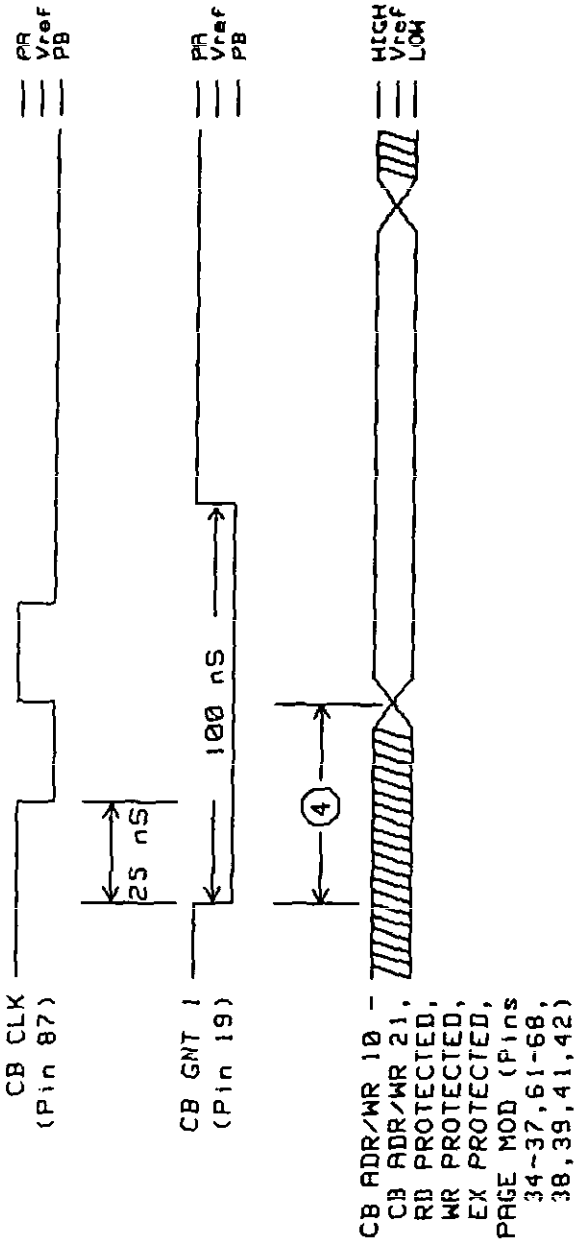


FIGURE 12b. Dynamic waveforms for table XVII.

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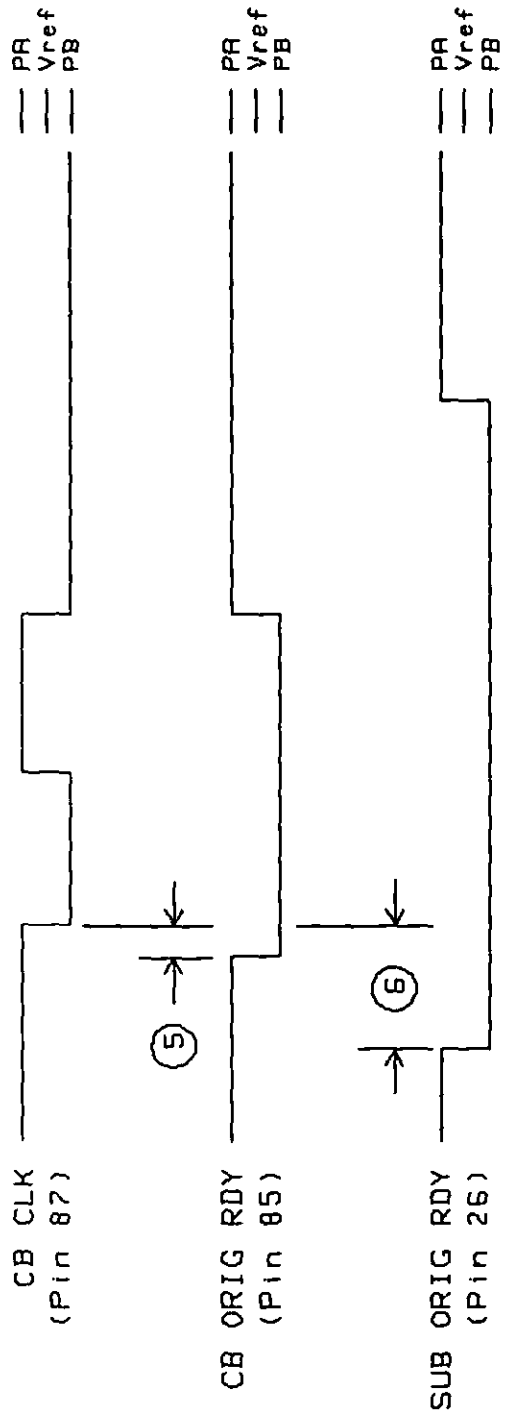


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	CB BSY IN	4	2	100	pF	1-4
R2	IOC CB WR	9	2	75	pF	1-4
R3	WR 15-12 IN	13	2	75	pF	1-4
R4	WR 11-0 IN	15	4	150	pF	1-4
R5	CB FCTN 0	16	2	75	pF	1-4
R6	CB FCTN 1	17				
R7	CB GNT 1	19	2	75	pF	1-4
R8	CLR IN	20	6	125	pF	1-4
R9	RD PAGE IN	21	2	75	pF	1-4
R10	ENBL BSY IN	23	4	100	pF	1-4
R11	CB FCTN 3	24	2	75	pF	1-4
R12	SUB ORIG RDY	26	2	75	pF	1-4
R13	IOC COMM REF	27	2	50	pF	1-4
R14	IOC WR REF	28	2	75	pF	1-4
R15	IOC RD REF	29				
R16	ADRS TRM RDY	30				
R17	CB WR ENBL	32				
R18	ANY SUB GNT	70				
R19	CB NDRO SEL	71	2	75	pF	1-4
R20	MACLR	72	6	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CB ADRS TO	73	2	75	pF	1-4
R22	CB DATA TO	74				
R23	ADRS 04	76				
R24	ADRS 05	77				
R25	ADRS 06	78				
R26	ADRS 07	79				
R27	ADRS 00	80				
R28	ADRS 01	82				
R29	ADRS 02	83				
R30	ADRS 03	84				
R31	CB ORIG RDY	85				
R32	CB FCTN 2	86				
R33	CB CLK	87				
R34	CB ADR/WR 08	88				
R35	CB ADR/WR 09	89	2	75	pF	1-4
R36	CB ADR/WR 00	93	4	75	pF	1-4
R37	CB ADR/WR 01	94				
R38	CB ADR/WR 02	95				
R39	CB ADR/WR 03	96				
R40	CB ADR/WR 04	97	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	CB ADR/WR 05	98	4	75	pF	1-4
R42	CB ADR/WR 06	99				
R43	CB ADR/WR 07	100	4	75	pF	1-4
Outputs						
R44	CB ADR/WR 18	34	2	75	pF	1-4
R45	CB ADR/WR 19	35				
R46	CB ADR/WR 20	36				
R47	CB ADR/WR 21	37				
R48	CB RD 08	43				
R49	CB RD 09	44				
R50	CB RD 10	45				
R51	CB RD 11	46				
R52	CB RD 12	47				
R53	CB RD 13	48				
R54	CB RD 14	49				
R55	CB RD 15	50				
R56	CB RD 07	52				
R57	CB RD 06	53				
R58	CB RD 05	54				
R59	CB RD 04	55				
R60	CB RD 03	56	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	CB RD 02	57	2	75	pF	1-4
R62	CB RD 01	58				
R63	CB RD 00	59				
R64	CB ADR/WR 17	61				
R65	CB ADR/WR 16	62	2	75	pF	1-4
Bidirectionals						
R66	CB ADR/WR 15	63	2	75	pF	1-4
R67	CB ADR/WR 14	64				
R68	CB ADR/WR 13	65				
R69	CB ADR/WR 12	66				
R70	CB ADR/WR 11	67				
R71	CB ADR/WR 10	68	2	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R71
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R71
Capacitance measurement	NA	± 4.9 pF	R1
	NA	± 3.65 pF	R2,R3
	NA	± 7.3 pF	R4
	NA	± 3.65 pF	R5-R7
	NA	± 5.95 pF	R8
	NA	± 3.65 pF	R9
	NA	± 4.8 pF	R10
	NA	± 3.65 pF	R11,R12
	NA	± 2.4 pF	R13
	NA	± 3.65 pF	R14-R19
	NA	± 3.45 pF	R20
	NA	± 3.65 pF	R21-R35
	NA	± 3.55 pF	R36-R43
	NA	± 3.65 pF	R44-R71

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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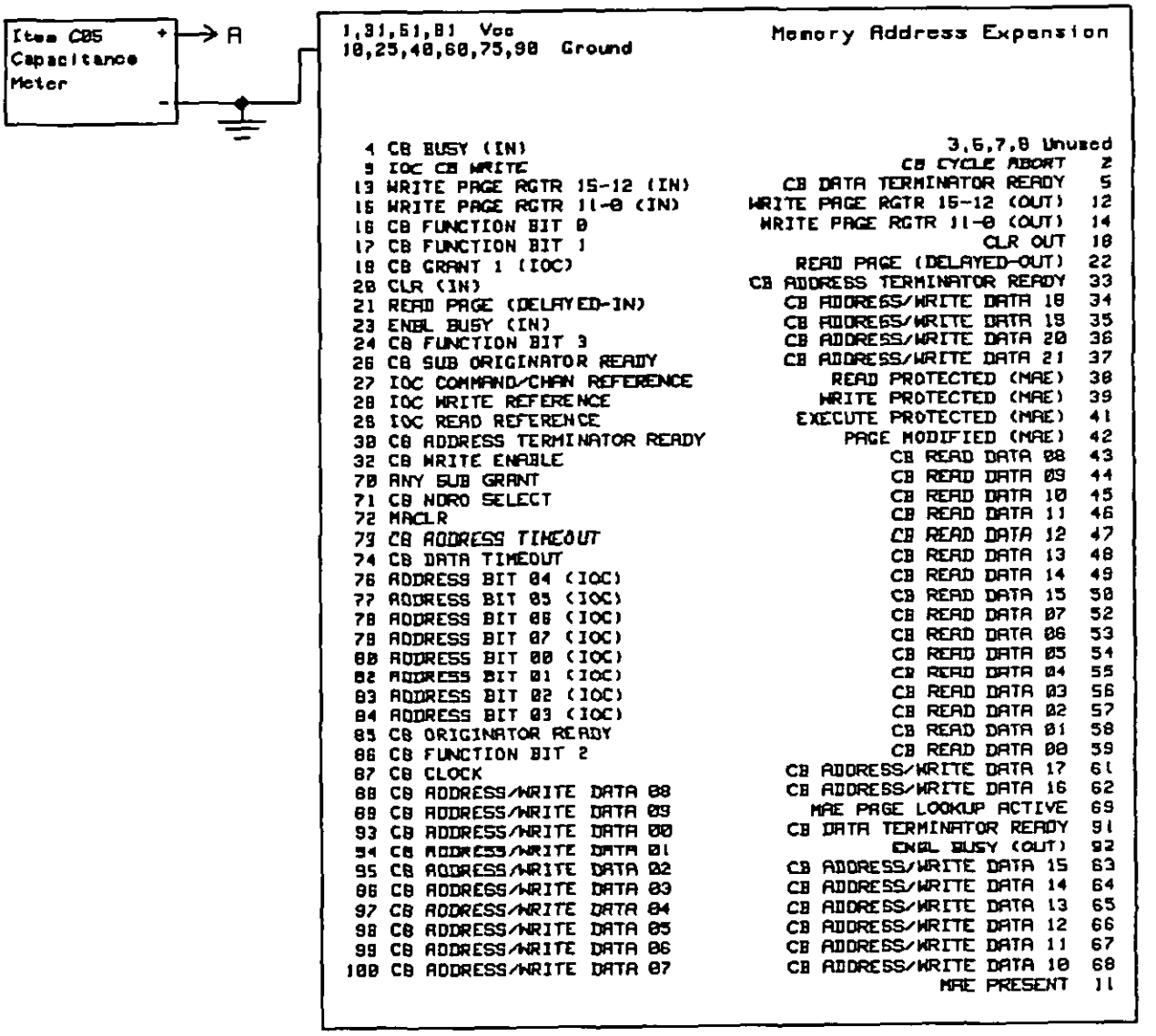


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latchup test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S	ANY SUB GNT	70	-18.0	-1.2	2.4	V	1-4

NOTES:

1. Input clamp diode latchup test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	S
Power supply current limit	2.36 A	NA	S
Forcing current, (Iin)	-18 mA	± 1.8 mA	S
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	S S
Input voltage measurement	NA	± 180 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

3. Apply pulse for a duration of 500 ± 20 nS.

4. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.36 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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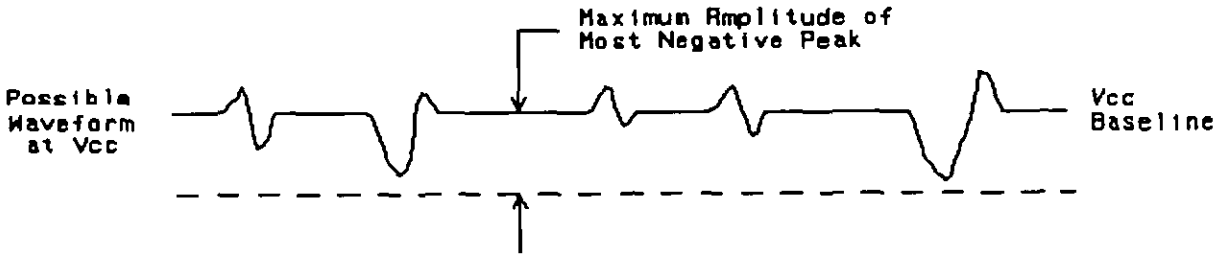
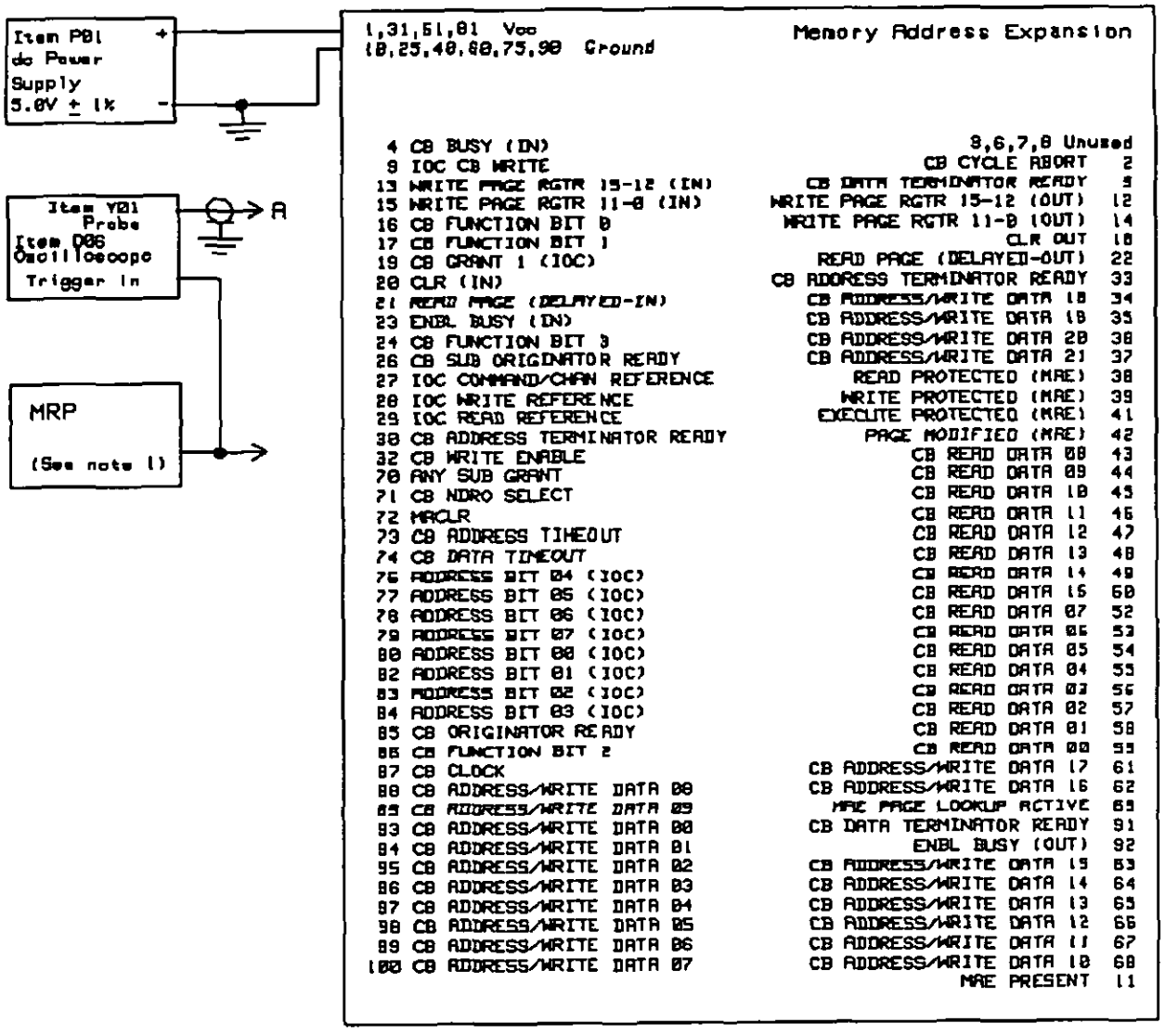


FIGURE 14. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XX.

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